

Markov Decision Process

Learning Outcomes

1. Understand the fundamentals of Markov Decision Processes (MDPs):

- Stochastic processes and the Markovian property.

2. Model real-world problems using MDPs:

- Identify states, decisions, transition probabilities, and costs/rewards in a given scenario (e.g., machine maintenance, inventory control).

3. Evaluate policies :

- Calculate steady-state probabilities for a given policy and compute the long-run expected average cost (or reward) of a policy.

4. Formulate an MDP as a linear programming (LP) problem to find optimal policies.

5. Contrast exhaustive enumeration (for small problems) with LP/policy improvement algorithms (for scalability).

Some key terms: Just to Recap

- **Stochastic process:** a process that evolves over time in a probabilistic manner.
- **Probabilistic manner** refers to a process or system that evolves or behaves according to probabilities rather than with absolute certainty.
- Each possible outcome has an associated likelihood (probability) of occurring
- **A stochastic process** is a mathematical concept used to describe a system or phenomenon that evolves over time with randomness or uncertainty.
- Example: the movement of a stock price over time

Markov Decision Processes (MDPs)

- **Goal:** Optimize performance by choosing actions in each state.
- **Components:**
 - States (e.g., machine conditions).
 - Decisions (e.g., keep as it is, repair, replace).
 - Costs (immediate and long-term).

Introduction

- What is a Markov Decision Process (MDP)?
 - A Markov Decision Process (MDP) is a mathematical framework used for decision-making in stochastic (random) environments where outcomes are partly random and partly under the control of a decision-maker.
- An MDP is a controlled Markov chain where an agent:
 - Observes the current state,
 - Chooses an action,
 - Receives a reward,
 - Transitions to a new state (with some probability).

Applications of MDP

- Healthcare
- Queueing Systems
- Highway Maintenance
- Key Objective
 - Minimizing expected average cost or optimizing performance through policy design.

Prototype Example

Manufacturer's machine deteriorates weekly

- Good as new : 0
- Minor deterioration (Operable) : 1
- Major deterioration (Operable) : 2
- Inoperable : 3

A machine at the core of a manufacturing process deteriorates over time. Let's take this example and understand the MDP

Historical Data and the Probability Matrix

- Based on historical data it show how the state of the machine evolves over time.
- Following is the probability matrix

State	0	1	2	3
0	0	$\frac{7}{8}$	$\frac{1}{16}$	$\frac{1}{16}$
1	0	$\frac{3}{4}$	$\frac{1}{8}$	$\frac{1}{8}$
2	0	0	$\frac{1}{2}$	$\frac{1}{2}$
3	0	0	0	1

Transition Probabilities

- Transition matrix based on historical data.

- Example:

- From State 0:

- $7/8 \rightarrow$ State 1,
 - $1/16 \rightarrow$ State 2,
 - $1/16 \rightarrow$ State 3.

State	0	1	2	3
0	0	$\frac{7}{8}$	$\frac{1}{16}$	$\frac{1}{16}$
1	0	$\frac{3}{4}$	$\frac{1}{8}$	$\frac{1}{8}$
2	0	0	$\frac{1}{2}$	$\frac{1}{2}$
3	0	0	0	1

- Absorbing State: State 3 (inoperable; must replace).

Costs and Decisions

- **Policy:**

- Replace when inoperable (state 3), otherwise do nothing.

- **Costs:**

- State 3: Replacement (\$4,000) + Lost production (\$2,000) = \$6,000.
 - Defective items: \$0 (state 0), \$1,000 (state 1), \$3,000 (state 2).

State	Expected Cost due to Defective Items, \$
0	0
1	1,000
2	3,000

Evaluation of the Maintenance Policy

- To evaluate this maintenance policy,
 - We should consider both the **immediate costs** incurred over the coming week and
 - the **subsequent costs** that result from having the system evolve in this way.
- To calculate this measure, we first derive the steady-state probabilities π_0 , π_1 , π_2 , and π_3 for this Markov chain.

Evaluating the Policy - (Steady-State Analysis)

- Steady-state equations:
 - $\pi_0 = \pi_3$
 - $\pi_1 = (7/8)\pi_0 + (3/4)\pi_1 + 0 \pi_2 + 0 \pi_3$
 - $\pi_2 = (1/16)\pi_0 + (1/8)\pi_1 + (1/2)\pi_2 + 0 \pi_3$
 - $\pi_3 = (1/16)\pi_0 + (1/8)\pi_1 + (1/2)\pi_2 + 0 \pi_3$
 - $\pi_0 + \pi_1 + \pi_2 + \pi_3 = 1$
- Solution: $\pi_0 = 2/13, \pi_1 = 7/13, \pi_2 = 2/13, \pi_3 = 2/13$.

Two step process:

Step 1: Simplify the Equations

Step 2: Use the Normalization Equation

Simplifying these equations

From **Equation 1**, we know:

$$\pi_0 = \pi_3$$

From **Equation 2**:

$$\pi_1 = \frac{7}{8}\pi_0 + \frac{3}{4}\pi_1$$

Subtract $\frac{3}{4}\pi_1$ from both sides:

$$\pi_1 - \frac{3}{4}\pi_1 = \frac{7}{8}\pi_0$$

$$\frac{1}{4}\pi_1 = \frac{7}{8}\pi_0$$

Multiply both sides by 4:

$$\pi_1 = \frac{28}{8}\pi_0 = \frac{7}{2}\pi_0$$

Equation 3

From **Equation 3**:

$$\pi_2 = \frac{1}{16}\pi_0 + \frac{1}{8}\pi_1 + \frac{1}{2}\pi_2$$

Subtract $\frac{1}{2}\pi_2$ from both sides:

$$\pi_2 - \frac{1}{2}\pi_2 = \frac{1}{16}\pi_0 + \frac{1}{8}\pi_1$$

$$\frac{1}{2}\pi_2 = \frac{1}{16}\pi_0 + \frac{1}{8}\pi_1$$

Multiply both sides by 2:

$$\pi_2 = \frac{1}{8}\pi_0 + \frac{1}{4}\pi_1$$

Substitute $\pi_1 = \frac{7}{2}\pi_0$:

$$\pi_2 = \frac{1}{8}\pi_0 + \frac{1}{4} \cdot \frac{7}{2}\pi_0 = \frac{1}{8}\pi_0 + \frac{7}{8}\pi_0 = \pi_0$$

Equation 4

$$\pi_3 = \frac{1}{16}\pi_0 + \frac{1}{8}\pi_1 + \frac{1}{2}\pi_2.$$

But we already have $\pi_3 = \pi_0$ (from Equation 1), and we've found $\pi_2 = \pi_0$. Substituting $\pi_1 = \frac{7}{2}\pi_0$ and $\pi_2 = \pi_0$:

$$\pi_0 = \frac{1}{16}\pi_0 + \frac{1}{8} \cdot \frac{7}{2}\pi_0 + \frac{1}{2}\pi_0$$

$$\pi_0 = \frac{1}{16}\pi_0 + \frac{7}{16}\pi_0 + \frac{8}{16}\pi_0$$

$$\pi_0 = \frac{16}{16}\pi_0$$

This simplifies to $\pi_0 = \pi_0$, which is consistent but doesn't give new information.

Step 2: Use the Normalization Equation

- $\pi_0 + \pi_1 + \pi_2 + \pi_3 = 1$
- Substitute $\pi_1 = \frac{7}{2}\pi_0$, $\pi_2 = \pi_0$, and $\pi_3 = \pi_0$:

$$\pi_0 + \frac{7}{2}\pi_0 + \pi_0 + \pi_0 = 1$$

- Convert all terms to have denominator 2:

$$\frac{2}{2}\pi_0 + \frac{7}{2}\pi_0 + \frac{4}{2}\pi_0 = 1$$

$$\frac{13}{2}\pi_0 = 1$$

$$\pi_0 = \frac{2}{13}$$

Finding other probabilities

- Using the value of π_0 calculating the values of π_1 , π_2 and π_3

$$\pi_1 = \frac{7}{2} \pi_0 = \frac{7}{2} \cdot \frac{2}{13} = \frac{7}{13}$$

$$\pi_2 = \pi_0 = \frac{2}{13}$$

$$\pi_3 = \pi_0 = \frac{2}{13}$$

Calculating the Cost Data of Prototype Example

TABLE 19.1 Cost data for the prototype example

Decision	State	Expected Cost due to Producing Defective Items, \$	Maintenance Cost, \$	Cost (Lost Profit) of Lost Production, \$	Total Cost per Week, \$
1. Do nothing	0	0	0	0	0
	1	1,000	0	0	1,000
	2	3,000	0	0	3,000
2. Overhaul	2	0	2,000	2,000	4,000
3. Replace	1, 2, 3	0	4,000	2,000	6,000

- Using the values in the previous slides, the (long-run) expected average cost per week for this maintenance policy is

$$0\pi_0 + 1,000\pi_1 + 3,000\pi_2 + 6,000\pi_3 = \frac{25,000}{13} = \$1,923.08.$$

Expected Average Cost (Long-Run Expected Cost)

- Formula: $E(C) = 0\pi_0 + 1,000\pi_1 + 3,000\pi_2 + 6,000\pi_3$
- Calculation:
$$= 0(2/13) + 1,000(7/13) + 3,000(2/13) + 6,000(2/13)$$
$$= 25,000/13 \approx \$1,923.08 \text{ per week.}$$
- Question: Is this the optimal policy?

- There are also other maintenance policies that should be considered and compared with this one.

Decision	Action	Relevant States
1	Do nothing	0, 1, 2
2	Overhaul (return system to state 1)	2
3	Replace (return system to state 0)	1, 2, 3

Alternative Decisions

- Exploring other options
- Decisions:
 - Do nothing (states 0, 1, 2).
 - Overhaul (state 2): \$2,000 maintenance + \$2,000 lost production = \$4,000, returns to state 1.
 - Replace (states 1, 2, 3): \$6,000, returns to state 0.
- Goal: Find the optimal policy using MDPs.

Next Steps

- Section 19.2: Model formulation for MDPs.
- Section 19.3: Linear programming to find optimal policy.
 - Objective: Minimize long-run expected average cost.

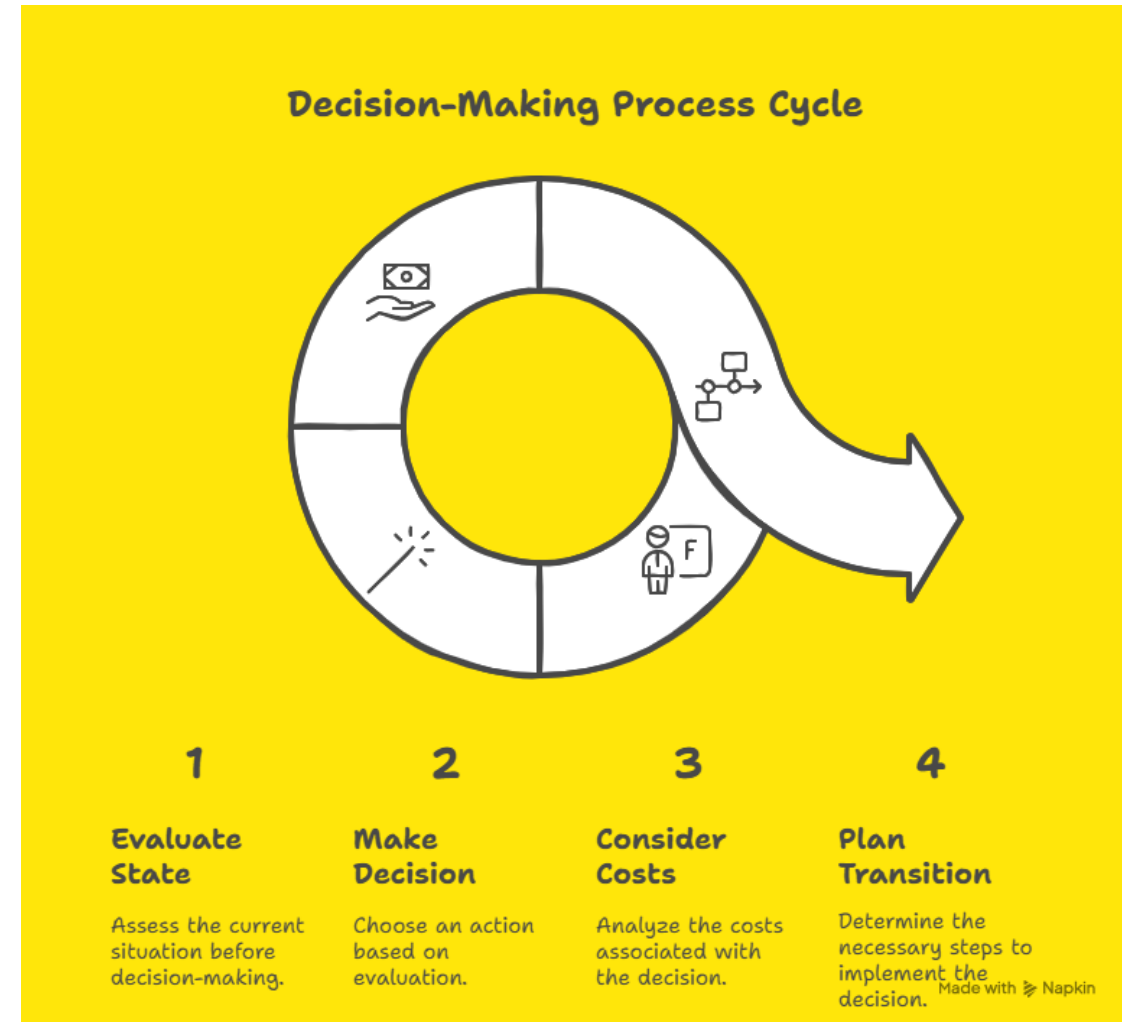
Section 19.2 - A Model for Markov Decision Process

Overview of the Section

- Formulate the MDP model for discrete-time Markov chains.
- Objective: Minimize long-run expected average cost per unit time.
- Define states, decisions, costs, and transition probabilities.
- Solve the prototype example using exhaustive enumeration.

MDP Model Components

- Core Elements of the MDP Model
 - States: Observed state i ($0, 1, \dots, M$) after each transition.
 - Decision Epoch: Choose action k from K options ($1, 2, \dots, K$).
 - Costs: Expected immediate cost C_{ik} for state i , decision k .
 - Transition Probabilities: $p_{ij}(k)$ – probability of moving from state i to j under decision k .



Assumptions and Policies

- Stationary and Deterministic Policies
- Stationary: Same decision rule for state i regardless of time t .
- Deterministic: One specific decision d_i chosen for each state i .
- Policy R : Set of decisions $\{d_0, d_1, \dots, d_M\} \rightarrow$ all set of possible actions available in states R
- Example: Prototype uses decisions 1 (do nothing), 2 (overhaul), 3 (replace).
- $R = \{d_0, d_1, \dots, d_M\}$.

Example Policy

- Policy: $(d_0, d_1, d_2, d_3) = (1, 1, 1, 3)$
 - States 0, 1, 2: Decision 1 (Do nothing).
 - State 3: Decision 3 (Replace)
- Markovian Property:
 - Future depends only on the *current state* and *decision* (not past history).

Policy Characteristics

1. Stationary Policy:

- Decision rule is fixed for each state (doesn't change over time).

2. Deterministic Policy:

- One specific decision is always chosen in a given state.

- Notation:

- R = A specific policy.
- $d_i(R)$ = Decision for state i under policy R .
- Possible Decisions: Only 1, 2, or 3 for any state.

Prototype Example Policies

- Policies for the Machine Maintenance Example
 - R_a : Replace in state 3 (1, 1, 1, 3).
 - R_b : Replace in state 3, overhaul in state 2 (1, 1, 2, 3).
 - R_c : Replace in states 2 and 3 (1, 1, 3, 3).
 - R_d : Replace in states 1, 2, and 3 (1, 3, 3, 3).

Policy	Verbal Description	$d_0(R)$	$d_1(R)$	$d_2(R)$	$d_3(R)$
R_a	Replace in state 3	1	1	1	3
R_b	Replace in state 3, overhaul in state 2	1	1	2	3
R_c	Replace in states 2 and 3	1	1	3	3
R_d	Replace in states 1, 2, and 3	1	3	3	3

Transition Matrices for Policies

- Transition Matrices for Each Policy

State	R_σ			
	0	1	2	3
0	0	$\frac{7}{8}$	$\frac{1}{16}$	$\frac{1}{16}$
1	0	$\frac{3}{4}$	$\frac{1}{8}$	$\frac{1}{8}$
2	0	0	$\frac{1}{2}$	$\frac{1}{2}$
3	1	0	0	0

Practice Assignment

Transition Matrices for Policies

State	R_ϕ			
	0	1	2	3
0	0	$\frac{7}{8}$	$\frac{1}{16}$	$\frac{1}{16}$
1	0	$\frac{3}{4}$	$\frac{1}{8}$	$\frac{1}{8}$
2	0	1	0	0
3	1	0	0	0

Practice Assignment

Transition Matrices for Policies

State	R_c			
	0	1	2	3
0	0	$\frac{7}{8}$	$\frac{1}{16}$	$\frac{1}{16}$
1	0	$\frac{3}{4}$	$\frac{1}{8}$	$\frac{1}{8}$
2	1	0	0	0
3	1	0	0	0

State	R_d			
	0	1	2	3
0	0	$\frac{7}{8}$	$\frac{1}{16}$	$\frac{1}{16}$
1	1	0	0	0
2	1	0	0	0
3	1	0	0	0

Cost Data Recap

- Costs for Decisions in Each State

TABLE 19.1 Cost data for the prototype example

Decision	State	Expected Cost due to Producing Defective Items, \$	Maintenance Cost, \$	Cost (Lost Profit) of Lost Production, \$	Total Cost per Week, \$
1. Do nothing	0	0	0	0	0
	1	1,000	0	0	1,000
	2	3,000	0	0	3,000
2. Overhaul	2	0	2,000	2,000	4,000
3. Replace	1, 2, 3	0	4,000	2,000	6,000

Expected Average Cost Formula

- Calculating Long-Run Expected Cost
- $E(C) = \sum_{i=0}^M C_{i,di} (R)^{\pi_i}$.
- π_i : Steady-state probability of state i under policy R .
- Solve steady-state equations for each policy's transition matrix.

Exhaustive Enumeration Results

- Comparing Policies for the Prototype

Policy	$(\pi_0, \pi_1, \pi_2, \pi_3)$	$E(C)$, in Thousands of Dollars
R_a	$\left(\frac{2}{13}, \frac{7}{13}, \frac{2}{13}, \frac{2}{13}\right)$	$\frac{1}{13}[2(0) + 7(1) + 2(3) + 2(6)] = \frac{25}{13} = \$1,923$
R_b	$\left(\frac{2}{21}, \frac{5}{7}, \frac{2}{21}, \frac{2}{21}\right)$	$\frac{1}{21}[2(0) + 15(1) + 2(4) + 2(6)] = \frac{35}{21} = \$1,667 \quad \leftarrow \text{Minimum}$
R_c	$\left(\frac{2}{11}, \frac{7}{11}, \frac{1}{11}, \frac{1}{11}\right)$	$\frac{1}{11}[2(0) + 7(1) + 1(6) + 1(6)] = \frac{19}{11} = \$1,727$
R_d	$\left(\frac{1}{2}, \frac{7}{16}, \frac{1}{32}, \frac{1}{32}\right)$	$\frac{1}{32}[16(0) + 14(6) + 1(6) + 1(6)] = \frac{96}{32} = \$3,000$

Optimal Policy Insight

- Optimal Policy: R_b
- Decisions: Do nothing (0, 1), overhaul (2), replace (3).
- $E(C) = \$1,667$ per week – lowest cost.
- Balances maintenance costs with production losses.
- Exhaustive enumeration works for small problems.

Limitations and Next Step

- Limitation: Infeasible for large numbers of states/policies.
- Next: Section 19.3 – Linear programming for efficiency.

Section 19.3 - Linear Programming and Optimal Policies

From Probabilities to Optimization

Overview of Section 19.3

- Why Linear Programming (LP) for MDPs?
 - Problem: Exhaustive enumeration is infeasible for large state/decision spaces.
 - Solution: LP efficiently finds optimal policies by reformulating MDP constraints.
- Goals of Linear Programming in MDPs
 - Efficiently find optimal policies for large MDP problems.
 - Move beyond exhaustive enumeration (Sec. 19.2).
 - Use LP to minimize long-run expected average cost.
 - Apply to prototype example for validation.

Why Use Linear Programming?

Challenges with Exhaustive Enumeration:

- Impractical for large state/decision spaces.
- Computationally expensive (e.g., 10 states $\times$ 5 decisions = 5^{10} policies!).

Advantages of LP:

- Efficiently finds optimal policies.
- Guarantees deterministic solutions.
- Uses standard optimization tools (Excel, Python, etc.).

Recap of MDP Components

1. **States** (e.g., machine conditions: 0, 1, 2, 3).
2. **Decisions/Actions** (e.g., do nothing (1), overhaul (2), replace(3)).
3. **Immediate Costs** (C_{ik}) and **Transition Probabilities** ($p_{ij}(k)$).
4. **Policy**: Rule mapping states to actions.

Policy Representation (Deterministic vs. Randomized Policies)

- **Deterministic Policy:**

- Fixed decision for each state (e.g., always replace in state 3).
- Represented by $D_{ik}=0$ or 1.

- **Randomized Policy:**

- Probabilistic decision for each state (e.g., replace with probability 1/2).
- Represented by $0 \leq D_{ik} \leq 1$.

LP Formulation

- Convert integer constraints ($D_{ik}=0$ or 1) into continuous variables for LP.

LP Formulation

- Goal: Minimize long-run expected average cost per unit time.

Variables:

- y_{ik} : Steady-state probability of being in state i and choosing action k .
- C_{ik} : The immediate expected cost incurred when in state i and taking decision k .
- Objective Function (Minimize expected average cost):

$$Z = \sum_{i=0}^M \sum_{k=1}^K C_{ik} y_{ik}$$

Example from Prototype Problem

- **States:** $i \in \{0,1,2,3\}$ (machine condition).
- **Decisions:** $k \in \{1(\text{do nothing}), 2(\text{overhaul}), 3(\text{replace})\}$.
- **Costs (C_{ik}):**
 - $C_{01} = 0$ (no cost if machine is good and left alone).
 - $C_{11} = 1000$ (cost if machine is in minor deterioration and left as it is).
 - $C_{23} = 6000$ (cost of replacing a majorly deteriorated machine).

The objective function becomes:

$$Z = 0 \cdot y_{01} + 1000 \cdot y_{11} + 6000 \cdot y_{13} + 3000 \cdot y_{21} + 4000 \cdot y_{22} + 6000 \cdot y_{23} + 6000 \cdot y_{33}$$

Understanding cost (the immediate expected cost)

State (i)	Decision (k)	Total Cost per Week (\$)	
0	1 (Do nothing)	0	
1	1 (Do nothing)	1,000	
1	3 (Replace)	6,000	Do nothing
2	1 (Do nothing)	3,000	but till there is
2	2 (Overhaul)	4,000	a cost, why?
2	3 (Replace)	6,000	
3	3 (Replace)	6,000	

Do nothing but till there is a cost - Reasons

- This cost are not direct maintenance cost but there are associated with the hidden operational costs such as
 - Lower-quality output – more defective products).
 - Lost productivity due to machine inefficiency.
- This reflects the opportunity cost

Decisions and immediate cost in state 2

Decision (k)	Action	Immediate Cost (C_{2k})	Explanation
1	Do nothing	\$3,000	Defective items cost.
2	Overhaul	\$4,000	2,000(overhaul)+2,000(overhaul)+2,000 (lost production).
3	Replace	\$6,000	4,000(replacement)+4,000(replacement)+2,000 (lost production).

Constraints

1. Probability Constraints

- Total probability must sum to 1:

$$\sum_{i=0}^M \sum_{k=1}^K y_{ik} = 1$$

2. Transition Balance Constraints

- For each state j , inflow = outflow:

$$\sum_{k=1}^K y_{jk} = \sum_{i=0}^M \sum_{k=1}^K y_{ik} p_{ij}(k)$$

• 3. Non-Negativity Constraints

$$y_{ik} \geq 0 \quad \forall i, k$$

Constraints Explanation

- First constraints: Total probability must be equal to 1.
- The **second set of constraints (transition balance constraints)** ensures that the **steady-state probabilities** of the Markov chain are **consistent with the system's dynamics**.
- More probability flows **into** a state than **out of it** → Leads to probability > 1 (make no sense).
- More probability flows **out of** a state than **into it** → Leads to probability < 0 (invalid).

Taking R_b example

State	R_b			
	0	1	2	3
0	0	$\frac{7}{8}$	$\frac{1}{16}$	$\frac{1}{16}$
1	0	$\frac{3}{4}$	$\frac{1}{8}$	$\frac{1}{8}$
2	0	1	0	0
3	1	0	0	0

Transition Balance Constraint Explained

- **Purpose:**

Ensures inflow = outflow for each state in steady-state.

- **Equation:**

$$\sum_k y_{jk} = \sum_i \sum_k y_{ik} p_{ij}(k)$$

- **Left Side:** Probability of being in state j .

- **Right Side:** Probability of transitioning into j from all states i .

- **Example (State 1 in Policy *Rb*):** $y_{11} = \frac{7}{8}y_{01} + \frac{3}{4}y_{11} + 1 \cdot y_{22}$

Transition Probabilities (under decision k=1, "do nothing"):

From \ To	0	1	2	3
0	0	$7/8$	$1/16$	$1/16$
1	0	$3/4$	$1/8$	$1/8$
2	0	1	0	0
3	1	0	0	0

LP Setup for Prototype Example

Objective:

Minimize $Z = 1000y_{11} + 6000y_{13} + 3000y_{21} + 4000y_{22} + 6000y_{23} + 6000y_{33}$.

Constraints:

1. $y_{01} + y_{11} + \cdots + y_{33} = 1$.
2. **State 0:** $y_{01} = y_{33}$.
3. **State 1:** $y_{11} = \frac{7}{8}y_{01} + \frac{3}{4}y_{11} + y_{22}$.
4. **State 2:** $y_{22} = \frac{1}{16}y_{01} + \frac{1}{8}y_{11}$.
5. **State 3:** $y_{33} = \frac{1}{16}y_{01} + \frac{1}{8}y_{11}$.

Transition Balance Constraint

Inflow rate (transitions into the state) = **Outflow rate** (transitions out of the state)

Transition Matrix for *Rb* - Recap

From \ To	State 0	State 1	State 2	State 3
State 0	0	7/8	1/16	1/16
State 1	0	3/4	1/8	1/8
State 2	0	1 (due to overhaul)	0	0
State 3	1 (due to replace)	0	0	0

Summary Table for Policy RbRb

State	Inflow Sources	Outflow Destinations	Balance Equation
0	State 3 (y_{33})	States 1,2,3 (y_{01})	$y_{01} = y_{33}$
1	States 0 ($\frac{7}{8}y_{01}$), 1 ($\frac{3}{4}y_{11}$), 2 (y_{22})	Self, States 2,3 (y_{11})	$y_{11} = \frac{7}{8}y_{01} + \frac{3}{4}y_{11} + y_{22}$
2	States 0 ($\frac{1}{16}y_{01}$), 1 ($\frac{1}{8}y_{11}$)	State 1 (y_{22})	$y_{22} = \frac{1}{16}y_{01} + \frac{1}{8}y_{11}$
3	States 0 ($\frac{1}{16}y_{01}$), 1 ($\frac{1}{8}y_{11}$)	State 0 (y_{33})	$y_{33} = \frac{1}{16}y_{01} + \frac{1}{8}y_{11}$

Markov Decision Processes in Practice

Why MDPs Matter in Practice

- Optimizes **sequential decision-making** under uncertainty.
- Balances immediate vs. long-term costs/rewards.
- Used in healthcare, finance, logistics, and more.

Example Application Areas

1. Healthcare

- *Chronic disease management*: Optimize screening/treatment schedules.

2. Transportation

- *Traffic light control*: Minimize wait times using real-time queue data.

3. Maintenance & Operations

- *Machine maintenance*: Replace/repair decisions (Prototype example).

4. Finance

- *Portfolio management*: Adjust investments based on market states.

Thanks